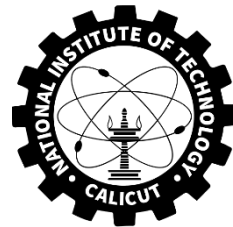


Quality, Reliability & Maintenance ME3327E



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Control Charts for the Mean and Standard Deviation



The sample standard deviation is given by

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}} = \sqrt{\frac{\sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2 / n}{n-1}}$$

If the population distribution of a quality characteristic is normal with a population standard deviation denoted by σ , the mean and standard deviation of the sample standard deviation are given by

$$E(s) = c_4 \sigma$$

$$\sigma_s = \sigma \sqrt{1 - c_4^2}$$

$$c_4 = \left[\frac{2}{n-1} \right]^{1/2} \frac{\Gamma(n/2)}{\Gamma[(n-1)/2]}$$

Control Chart Patterns and Corrective

APPENDIX A-7 Factors for Computing Centerline and Three-Sigma Control Limits

Observations in Sample, n	$\bar{X}$ -Charts			s -Charts						R -Charts						
	Factors for Control Limits			Factors for Centerline		Factors for Control Limits				Factors for Centerline		Factors for Control Limits				
	A	A ₂	A ₃	c_4	$1/c_4$	B ₃	B ₄	B ₅	B ₆	d_2	$1/d_2$	d_3	D ₁	D ₂	D ₃	D ₄
2	2.121	1.880	2.659	0.7979	1.2533	0	3.267	0	2.606	1.128	0.8865	0.853	0	3.686	0	3.267
3	1.732	1.023	1.954	0.8862	1.1284	0	2.568	0	2.276	1.693	0.5907	0.888	0	4.358	0	2.574
4	1.500	0.729	1.628	0.9213	1.0854	0	2.266	0	2.088	2.059	0.4857	0.880	0	4.698	0	2.282
5	1.342	0.577	1.427	0.9400	1.0638	0	2.089	0	1.964	2.326	0.4299	0.864	0	4.918	0	2.114
6	1.225	0.483	1.287	0.9515	1.0510	0.030	1.970	0.029	1.874	2.534	0.3946	0.848	0	5.078	0	2.004
7	1.134	0.419	1.182	0.9594	1.0423	0.118	1.882	0.113	1.806	2.704	0.3698	0.833	0.204	5.204	0.076	1.924
8	1.061	0.373	1.099	0.9650	1.0363	0.185	1.815	0.179	1.751	2.847	0.3512	0.820	0.388	5.306	0.136	1.864
9	1.000	0.337	1.032	0.9693	1.0317	0.239	1.761	0.232	1.707	2.970	0.3367	0.808	0.547	5.393	0.184	1.816
10	0.949	0.308	0.975	0.9727	1.0281	0.284	1.716	0.276	1.669	3.078	0.3249	0.797	0.687	5.469	0.223	1.777
11	0.905	0.285	0.927	0.9754	1.0252	0.321	1.679	0.313	1.637	3.173	0.3152	0.787	0.811	5.535	0.256	1.744
12	0.866	0.266	0.886	0.9776	1.0229	0.354	1.646	0.346	1.610	3.258	0.3069	0.778	0.922	5.594	0.283	1.717
13	0.832	0.249	0.850	0.9794	1.0210	0.382	1.618	0.374	1.585	3.336	0.2998	0.770	1.025	5.647	0.307	1.693
14	0.802	0.235	0.817	0.9810	1.0194	0.406	1.594	0.399	1.563	3.407	0.2935	0.763	1.118	5.696	0.328	1.672
15	0.775	0.223	0.789	0.9823	1.0180	0.428	1.572	0.421	1.544	3.472	0.2880	0.756	1.203	5.741	0.347	1.653
16	0.750	0.212	0.763	0.9835	1.0168	0.448	1.552	0.440	1.526	3.532	0.2831	0.750	1.282	5.782	0.363	1.637
17	0.728	0.203	0.739	0.9845	1.0157	0.466	1.534	0.458	1.511	3.588	0.2787	0.744	1.356	5.820	0.378	1.622
18	0.707	0.194	0.718	0.9854	1.0148	0.482	1.518	0.475	1.496	3.640	0.2747	0.739	1.424	5.856	0.391	1.608
19	0.688	0.187	0.698	0.9862	1.0140	0.497	1.503	0.490	1.483	3.689	0.2711	0.734	1.487	5.891	0.403	1.597
20	0.671	0.180	0.680	0.9869	1.0133	0.510	1.490	0.504	1.470	3.735	0.2677	0.729	1.549	5.921	0.415	1.585
21	0.655	0.173	0.663	0.9876	1.0126	0.523	1.477	0.516	1.459	3.778	0.2647	0.724	1.605	5.951	0.425	1.575
22	0.640	0.167	0.647	0.9882	1.0119	0.534	1.466	0.528	1.448	3.819	0.2618	0.720	1.659	5.979	0.434	1.566
23	0.626	0.162	0.633	0.9887	1.0114	0.545	1.455	0.539	1.438	3.858	0.2592	0.716	1.710	6.006	0.443	1.557
24	0.612	0.157	0.619	0.9892	1.0109	0.555	1.445	0.549	1.429	3.895	0.2567	0.712	1.759	6.031	0.451	1.548
25	0.600	0.153	0.606	0.9896	1.0105	0.565	1.435	0.559	1.420	3.931	0.2544	0.708	1.806	6.056	0.459	1.541

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Control Charts for the Mean and Standard Deviation

No Given Standards

The centerline of a standard deviation chart is

$$CL_s = \bar{s} = \frac{\sum_{i=1}^g s_i}{g}$$

$$UCL_s = B_4 \bar{s}$$

$$LCL_s = B_3 \bar{s}$$

The centerline of the chart for the mean $\bar{X}$ is given by

$$CL_{\bar{X}} = \bar{\bar{X}} = \frac{\sum_{i=1}^g \bar{X}_i}{g}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_3 \bar{s}$$

$$LCL_{\bar{X}} = \bar{\bar{X}} - A_3 \bar{s}$$

Control Charts for the Mean and Standard Deviation



Given Standards

The centerline of a standard deviation chart is

$$CL_s = c_4 \sigma_o$$

$$UCL_s = B_6 \sigma_o$$

$$LCL_s = B_5 \sigma_o$$

The centerline of the chart for the mean $\bar{X}$ is given by

$$CL_{\bar{X}} = \bar{X}_o$$

$$UCL_{\bar{X}} = \bar{X}_o + A \sigma_o$$

$$LCL_{\bar{X}} = \bar{X}_o - A \sigma_o$$

Control Charts for Individual Units

No Given Standards

The centerline of a moving range (MR) chart is

$$\begin{aligned}CL_{MR} &= \overline{MR} \\UCL_{MR} &= D_4 \overline{MR} \\LCL_{MR} &= D_3 \overline{MR}\end{aligned}$$

The centerline of the chart for the mean $\bar{X}$ is given by

$$\begin{aligned}CL_{\bar{X}} &= \bar{\bar{X}} \\UCL_{\bar{X}} &= \bar{\bar{X}} + 3 \frac{\overline{MR}}{d_2} \\LCL_{\bar{X}} &= \bar{\bar{X}} - 3 \frac{\overline{MR}}{d_2}\end{aligned}$$

Control Charts for Individual Units

Given Standards

The centerline of a moving range (MR) chart is

$$CL_{MR} = d_2 \sigma_o$$

$$UCL_{MR} = D_4 d_2 \sigma_o$$

$$LCL_{MR} = D_3 d_2 \sigma_o$$

The centerline of the chart for the mean $\bar{X}$ is given by

$$CL_{\bar{X}} = \bar{X}_o$$

$$UCL_{\bar{X}} = \bar{X}_o + 3\sigma_o$$

$$LCL_{\bar{X}} = \bar{X}_o - 3\sigma_o$$

Control Charts for Attributes

Definition of Attribute Data and Examples

Attribute data refers to quality characteristics that are counted or classified rather than measured on a continuous scale. An attribute describes something in terms of distinct categories or labels, usually as "conforming" (meets a standard) or "nonconforming" (fails to meet a standard). In statistics and quality control, attribute data is measured on a nominal scale—for example, pass/fail, accept/reject, or by counting the number of occurrences of a specific event.

Examples:

- Counting the number of faulty (nonconforming) light bulbs in a batch.
- Recording if a product passes or fails a final inspection.
- Tallying the number of errors or defects on a printed page.
- Classifying a food item as "acceptable" or "not acceptable" based on taste.
- Recording the number of insurance claims filed due to errors in policy processing.

Difference Between Attributes and Variables

Aspect	Attributes (Attribute Data)	Variables (Variable Data)
Definition	Qualitative, categorical; classified into discrete groups	Quantitative; measured on a numerical, continuous scale
Measurement Scale	Nominal (e.g., yes/no, defect/no defect); integer counts	Interval or ratio (e.g., length in cm, weight in kg)
Data Collected As	Counts or classifications (e.g., number of defects, pass/fail)	Actual measurements (e.g., thickness, volume, temperature)
Examples	Number of defective items, type of defect	Size, weight, temperature, pressure
Statistical Charts	Attribute control charts (e.g., p-chart, np-chart, c-chart, u-chart)	Variable control charts (e.g., X-bar, R chart, S chart)

Importance of Attribute Control Charts:

- Attribute control charts provide a cost-effective and simple method for monitoring processes where direct measurement is impractical or unnecessary.
- They help in summarizing overall quality at various organizational levels (e.g., operator, department, plant).
- Useful when decisions are based on compliance with specifications rather than degree of conformance.

Application of Attribute Control Charts in Quality Control



Key Applications:

- **Manufacturing:** Monitoring the proportion of defective parts, tracking the number of flaws in batches, or counting faulty units.
- **Service industry:** Recording customer complaints, transaction errors, or incorrect billing instances.
- **Healthcare:** Counting adverse events (e.g., infections, medication errors) or monitoring mortality/morbidity rates.
- **Logistics:** Proportion of on-time deliveries versus late deliveries.
- **Administrative processes:** Counting the number of application rejections, claims processed incorrectly, or forms returned for corrections.

Why Use Attribute Charts?

- They allow for overall assessment of process quality, especially when measuring every output dimension is impractical.
- They are straightforward for operators to understand and implement.
- Suitable for high-volume processes or when quick, binary (pass/fail) feedback is essential.
- Examples include p-charts (proportion nonconforming), np-charts (number nonconforming), c-charts (number of nonconformities), and u-charts (nonconformities per unit).

By deploying attribute control charts, organizations can efficiently monitor, identify, and address quality issues, thereby supporting ongoing process improvement and customer satisfaction.

Types of Control Charts for Attributes

Chart Type	Tracks	Key Distribution	When to Use
p-chart	Proportion nonconforming	Binomial	Variable/non-constant sample size
np-chart	Number nonconforming	Binomial	Constant sample size
c-chart	Number of nonconformities	Poisson	Defects per fixed area/unit, constant sample size
u-chart	Number of nonconformities per unit	Poisson	Variable area/sample size
U-chart	Weighted demerits per unit		Weighted severity of defects
g-chart	Cases/time between nonconformities	Exponential/Geometric	Highly conforming processes

Advantages of Attribute Control Charts

- **Simplicity of Use:** Attribute charts are easy to construct and interpret, making them accessible for operators and managers at all organizational levels.
- **Summarization:** They allow you to condense multiple quality characteristics or components into a single, overall quality indicator. For example, a product that must meet specifications for length, width, and height can be monitored with one attribute chart instead of three variable charts.
- **Cost and Resource Efficiency:** Ideal for cases where measuring each characteristic as a variable is too expensive, time-consuming, or resource-intensive. Attribute data collection often requires less specialized equipment—sometimes just a visual inspection or a go/no-go gage.
- **Broad Applicability:** Can be used for summarizing quality at higher organizational levels (e.g., plant, department, company), helping to quickly identify where problems are occurring without the need to track every individual process parameter.
- **Suitable for Certain Quality Characteristics:** Some quality characteristics (e.g., taste, visual appearance, presence of a scratch) are inherently categorical and can't be meaningfully measured as variables.

Disadvantages of Attribute Control Charts

- **Lack of Detailed Information:** Attribute charts provide only "conforming" or "nonconforming" data—they do not indicate by how much a measurement deviates from the target. This means they do not show the degree of variability or how close items are to specification limits.
- **Limited Diagnostic Power:** They do not provide insights into the mean or the spread of the distribution, so it's harder to diagnose the root cause of process shifts compared to variable charts (like $\bar{X}$ and R charts).
- **Slower Detection:** Attribute charts may not signal a process shift until out-of-spec items actually occur. Variable charts can sometimes warn of impending problems before any defectives are produced.
- **Require Larger Sample Sizes:** To achieve the same sensitivity as variable charts, attribute charts often require larger samples—especially problematic when inspection is costly or destructive.

Chart for Proportion Nonconforming: p-Chart

A chart for the proportion of nonconforming items (p-chart) is based on a binomial distribution. For a given sample, the proportion

nonconforming is defined as $\hat{p} = \frac{x}{n}$

Where x is the number of nonconforming items in the sample and n represents the sample size.

The distribution of the number of nonconforming items in a sample, as given by a binomial distribution, is

$$P(X = x) = \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x}$$

Given the mean and standard deviation of X , the mean of the sample proportion nonconforming is $E(\hat{p}) = p$

$$Var(\hat{p}) = \frac{p(1-p)}{n}$$

p-Chart: Construction and Interpretation

Step-by-step walkthrough:

- Define objective (plant, department, operator, etc.)
- Decide sample size and interval (criteria/guidelines) ($n\bar{p} \geq 5$)
- Record and organize sample data
- Calculate centerline and control limits (with formulae)
- Plot, interpret out-of-control signals, and revise limits
- Ongoing use and periodic review

Calculate centerline and control limits

No Standard Specified:

When no standard or target value of the proportion nonconforming is specified, it must be estimated from the sample information.

$$CL_p = \bar{p} = \frac{\sum_{i=1}^g \hat{p}_i}{g} = \frac{\sum_{i=1}^g x_i}{ng}$$

$$UCL_p = \bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$LCL_p = \bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

Calculate centerline and control limits

Standard Specified:

If the target value of the proportion of nonconforming items is known or specified, the centerline is selected as that target value. In other words, the centerline is given by

$$CL_p = p_o$$

$$UCL_p = p_o + 3 \sqrt{\frac{p_o(1 - p_o)}{n}}$$

$$LCL_p = p_o - 3 \sqrt{\frac{p_o(1 - p_o)}{n}}$$

Calculate the revised control limits

The revised centerline and control limits are given by

$$CL_p = \bar{p}_r$$

$$UCL_p = \bar{p}_r + 3 \sqrt{\frac{\bar{p}_r(1 - \bar{p}_r)}{n}}$$

$$LCL_p = \bar{p}_r - 3 \sqrt{\frac{\bar{p}_r(1 - \bar{p}_r)}{n}}$$

Variable Sample Size

There are many reasons why samples vary in size.

Control Limits for Individual Samples

Control limits can be constructed for individual samples. If no standard is given and the sample average proportion nonconforming is $\bar{p}$, the control limits for sample i with size n_i are

$$UCL_p = \bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

$$LCL_p = \bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

Variable Sample Size

Standardized Control Chart

Another approach to varying sample size is to construct a chart of normalized or standardized values of the proportion nonconforming.

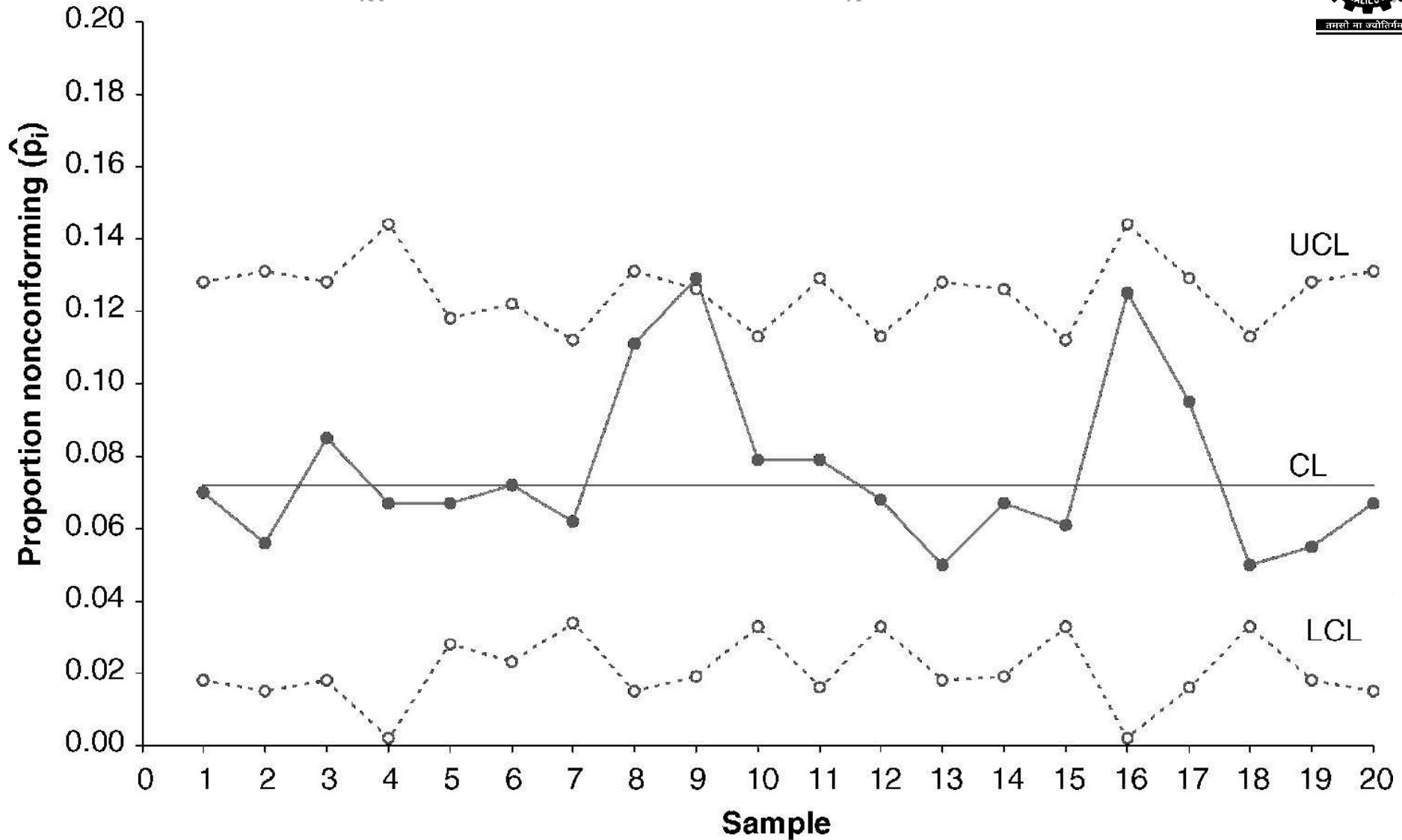
$$E(\hat{p}) = p$$
$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

respectively, where p represents the true process nonconformance rate and n is the sample size.

The standardized value of the proportion nonconforming for the i th sample may be expressed as

$$Z = \frac{\hat{p}_i - \bar{p}}{\sqrt{\bar{p}(1-\bar{p})/n_i}}$$

Variable Sample Size



Proportion-nonconforming chart for variable sample sizes.

Special Considerations for p-Charts

- Necessary Assumptions
- Observations Below the Lower Control Limit
- Comparison with a Specified Standard
- Impact of Design Specifications
- Information About Overall Quality Level

Chart for Number of Nonconforming Items: np-Chart



As an alternative to calculating the proportion nonconforming, we can count the number of nonconforming items in samples and use the count as the basis for the control chart.

No Standard Specified:

When no standard or target value is specified, it must be estimated from the sample information.

$$CL_{np} = n\bar{p} = \frac{\sum_{i=1}^g x_i}{g}$$

$$\sigma_{np} = \sqrt{n\bar{p}(1 - n\bar{p})}$$

$$UCL_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1 - n\bar{p})}$$

$$LCL_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1 - n\bar{p})}$$

Chart for Number of Nonconforming Items: np-Chart

Standard Specified:

If the target value is known or specified, the centerline is selected as that target value. In other words, the centerline is given by

$$CL_{np} = np_o$$

$$UCL_{np} = np_o + 3\sqrt{np_o(1 - np_o)}$$

$$LCL_{np} = np_o - 3\sqrt{np_o(1 - np_o)}$$

Chart for Number of Nonconformities: c-Chart



A *nonconformity* is defined as a quality characteristic that does not meet some specification. A nonconforming item has one or more nonconformities that make it nonfunctional. It is also possible for a product to have one or more nonconformities and still conform to standards. The p- and np-charts deal with nonconforming items. A c-chart is used to track the total number of nonconformities in samples of constant size. When the sample size varies, a u-chart is used to track the number of nonconformities per unit. In constructing c- and u-charts, the size of the sample is also referred to as the area of opportunity.

The occurrence of nonconformities is assumed to follow a *Poisson distribution*. If x represents the number of nonconformities in the sample unit and c is the mean, then the Poisson distribution yields

$$p(x) = \frac{e^{-c} c^x}{x!}$$

Chart for Number of Nonconformities: c-Chart



No Standard Specified:

The average number of nonconformities per sample unit is found from the sample observations and is denoted by $\bar{c}$. The centerline and control limits are

$$CL_c = \bar{c}$$

$$UCL_c = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL_c = \bar{c} - 3\sqrt{\bar{c}}$$

Standard Specified:

Let the specified goal for the number of nonconformities per sample unit be c_o . The centerline and control limits are then calculated from

$$CL_c = c_o$$

$$UCL_c = c_o + 3\sqrt{c_o}$$

$$LCL_c = c_o - 3\sqrt{c_o}$$

Chart for Number of Nonconformities Per Unit: u-Chart



A c-chart is used when the sample size is constant. If the area of opportunity changes from one sample to another, the centerline and control limits of a c-chart change as well. For situations in which the sample size varies, a u-chart is used. For companies that inspect all items produced or services rendered for the presence of nonconformities, the output per production run can vary because of fluctuating supplies of labor, machinery, and raw material; consequently, the number inspected per production run changes, thus causing varying sample sizes. When the sample size varies, a u-chart is constructed to monitor the number of nonconformities per unit. Even though the control limits change as the sample size varies, the centerline of a u-chart remains constant, which permits meaningful comparisons between the samples.

Chart for Number of Nonconformities Per Unit: u-Chart

Variable Sample Size and No Specified Standard:

When the sample size varies, the number of nonconformities per unit for the i th sample is given by

$$u_i = \frac{c_i}{n_i}$$

The average number of nonconformities per unit $\bar{u}$, which is also the centerline of a u-chart, is given by

$$\bar{u} = \frac{\sum_{i=1}^g c_i}{\sum_{i=1}^g n_i}$$

$$UCL_u = \bar{u} + 3 \sqrt{\frac{\bar{u}}{n_i}}$$

$$LCL_u = \bar{u} - 3 \sqrt{\frac{\bar{u}}{n_i}}$$

Chart for Demerits Per Unit: U-Chart

The regular c- and u-charts treat all types of nonconformities equally regardless of their degree of severity. Let's suppose that in inspecting computer monitors we find that one monitor has trouble retaining consistent color and a second monitor has five scratch marks on its surface. Using either a c- or u-chart, the relative importance of monitor 2's defects, in terms of the number of nonconformities, is five times as great as that of monitor 1. However, the single defect associated with monitor 1 is much more serious than monitor 2's scratch marks. An alternative approach assigns weights to nonconformities according to their relative degree of severity. This quality rating system, which rates demerits per unit and is called the U-chart, thus overcomes the deficiency of the c- and u-charts. These are often helpful in service applications.

Construction of a U-Chart

Suppose that we have four categories of nonconformities.

For a sample of size n , the total number of demerits is given by

$$D = w_1c_1 + w_2c_2 + w_3c_3 + w_4c_4$$

The demerits per unit for the sample are given by, $U = \frac{D}{n}$

The centerline of the U-chart is given by

$$\bar{U} = w_1\bar{u}_1 + w_2\bar{u}_2 + w_3\bar{u}_3 + w_4\bar{u}_4$$

The estimated standard deviation of U is given by

$$\hat{\sigma}_U = \sqrt{\frac{w_1^2\bar{u}_1 + w_2^2\bar{u}_2 + w_3^2\bar{u}_3 + w_4^2\bar{u}_4}{n}}$$

The control limits for the U-chart are given by

$$UCL_U = \bar{U} + 3\hat{\sigma}_U$$

$$LCL_U = \bar{U} - 3\hat{\sigma}_U$$

Charts for Highly Conforming Processes

Use of Exponential Distribution for Continuous Variables:

When the process under consideration is continuous, an alternative under the assumption that the occurrence of nonconformities conforms to a *Poisson* process with mean λ is to monitor the time or number of items required (Q) to observe exactly one non conformity.

Hence, the probability of observing Q units in order to observe a defect is $F(Q) = 1 - e^{-\lambda Q}, Q \geq 0$

The centerline and control limits are then calculated from

$$CL = -\frac{1}{\lambda} \ln \left(\frac{1}{2} \right) = \frac{0.6931}{\lambda}$$

$$LCL = -\frac{1}{\lambda} \ln \left(1 - \frac{\alpha}{2} \right)$$

$$UCL = -\frac{1}{\lambda} \ln \left(\frac{\alpha}{2} \right)$$

Charts for Highly Conforming Processes

Use of Geometric Distribution for Discrete Variables:

For a highly conforming process, non conforming items are few and far between. This implies that monitoring the number of nonconforming items will yield most observations with values of zero, which prevents the detection of a change in the nonconformance rate. An alternative is to monitor the number of items until a nonconforming item is found, sometimes referred to as the number of trials upto the first success (X). When a nonconforming item is found, the count starts a new. The discrete variable X , as defined, has a geometric distribution given by $P(X = x) = (1 - p)^{x-1}p$, $x = 1, 2, \dots$

Where p represents the probability of success (nonconforming item) on each trial. The mean and variance of X , are given by

$$E(X) = \frac{1}{p}, \text{Var}(X) = \frac{1 - p}{p^2}$$

Charts for Highly Conforming Processes

Use of Geometric Distribution for Discrete Variables :

The centerline and control limits are then calculated from

$$CL = \frac{1}{p}$$

$$LCL = \frac{\ln\left(1 - \frac{\alpha}{2}\right)}{\ln(1 - p)}$$

$$UCL = \frac{\ln\left(\frac{\alpha}{2}\right)}{\ln(1 - p)}$$

The probability of no nonconforming items in a sample of size n is

$$P(0 \text{ nonconforming items}) = (1 - p)^n$$